

Dot matrix OLED display module Manual

QT095B

1. FUNCTIONS & FEATURES

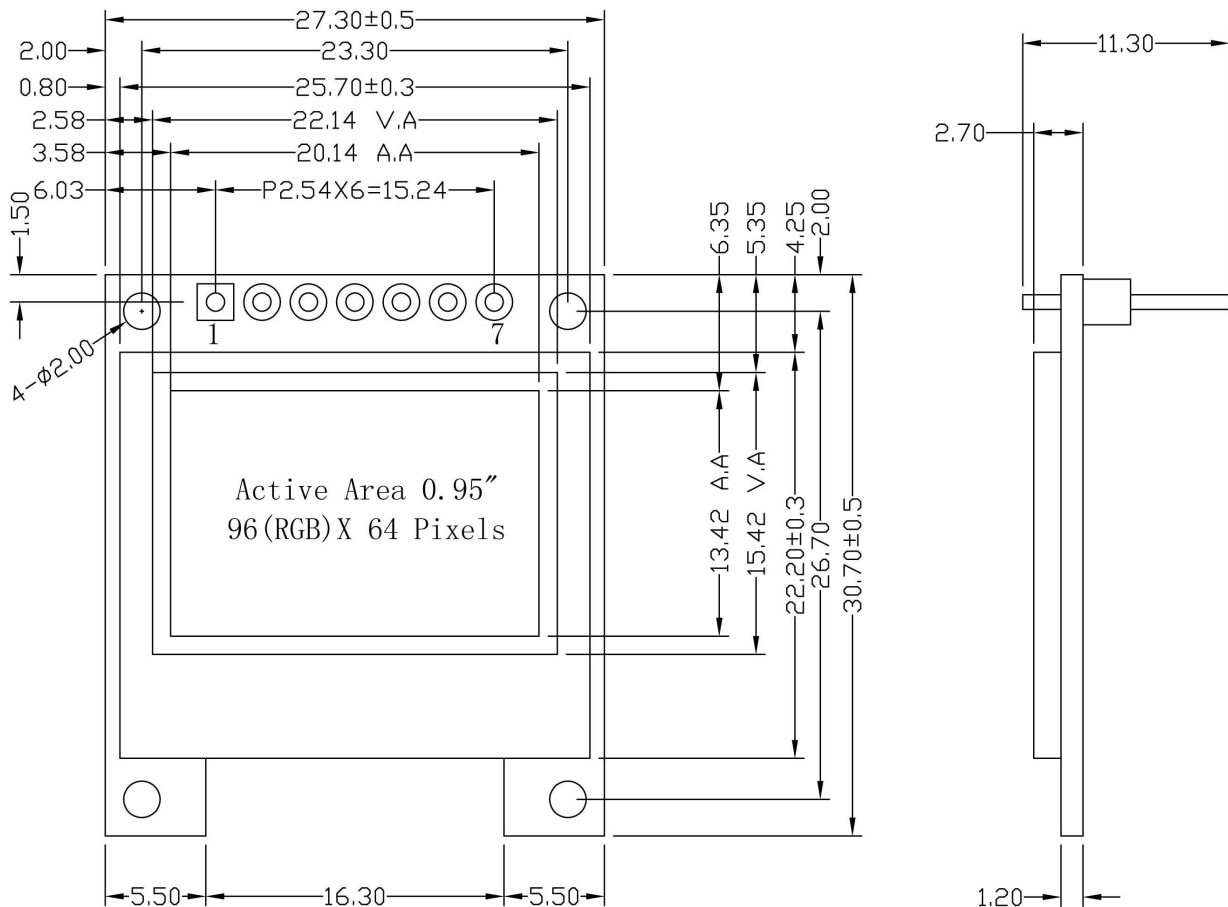
Features

- Display Mode: Passive Matrix OLED
- Number of Pixels: 96(RGB)×64 dots
- Display Color: 65,536 Colors
- Drive Duty: 1/64 Duty
- Driver IC: SSD1331
- 4SPI Interface

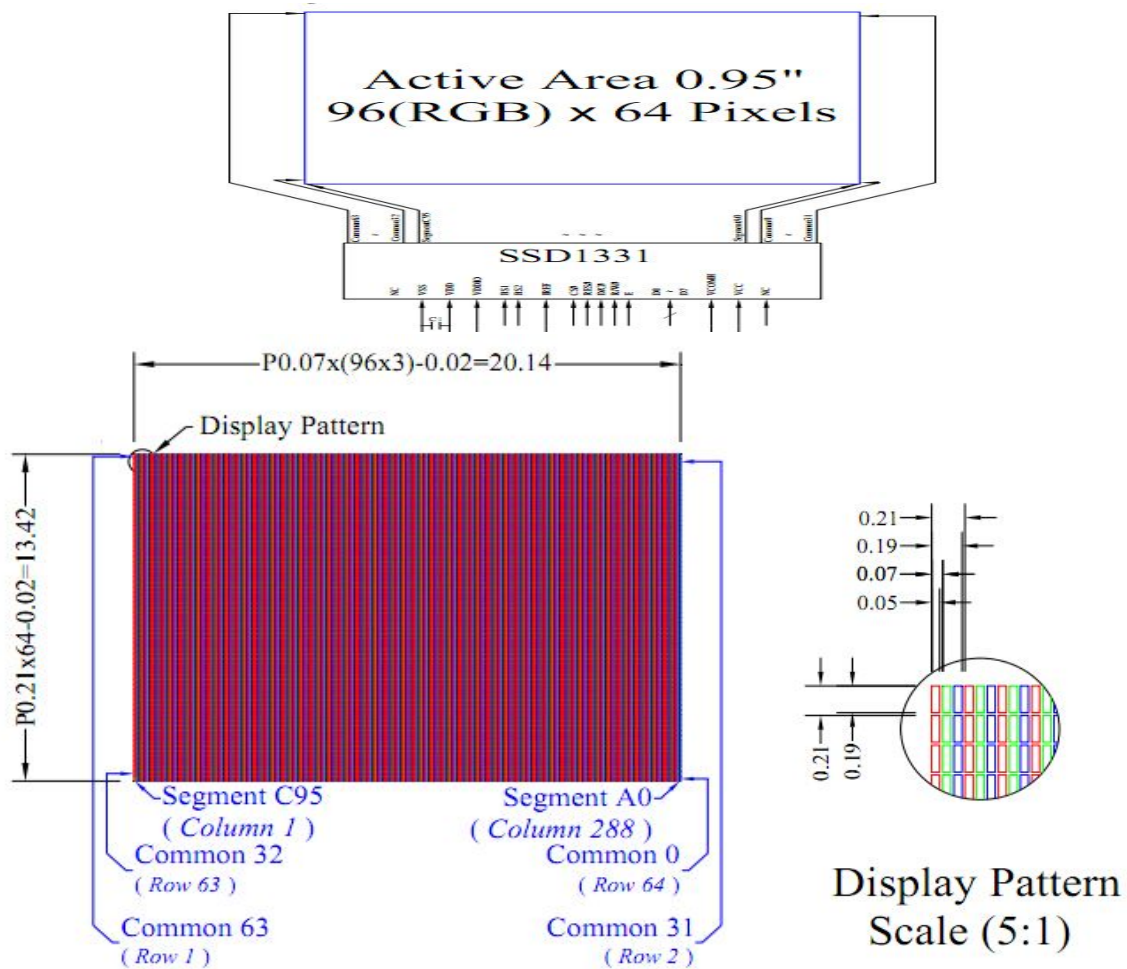
2. MECHANICAL SPECIFICATIONS

ITEM	SPECIFICATIONS	UNIT
Module Size	27.3L×30.7W×11.3(MAX) H	mm
View Area	22.14×15.42	mm
Effective Area	96 (RGB) ×64	dots
Pixel Size	0.05×0.19	mm
Pixel Pitch	0.07×0.21	mm

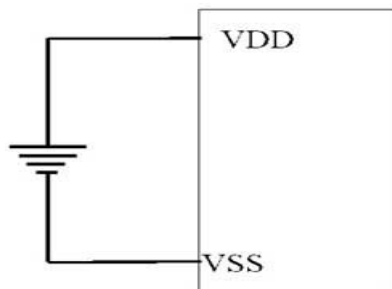
3. EXTERNAL DIMENSIONS



4. BLOCK DIAGRAM



5. POWER SUPPLY



6. PIN DESCRIPTION

ITEM	SYMBOL	LEVEL	FUNCTION
1	VSS	0V	Power Ground
2	VDD	+3.3V~5.0V	Power Supply For Logic
3	SCLK	H/L	Serial Clock signal
4	SDIN	H/L	Serial Data input signal
5	RES	H/L	Active LOW Reset signal
6	D/C	H/L	Data/Command control
7	CS	L	Chip select input. (active LOW).

7 . Absolute Maximum Ratings

Parameter	Symbol	Min	Max	Unit	Notes
Supply Voltage for Logic	Vdd	-0.3	4	V	1, 2
Supply Voltage for Display	Vcc	0	15	V	1, 2
Vcc Supply Current	Icc	-	25	mA	1, 2
Operating Temperature	Top	-30	70	°C	-
Storage Temperature	Tst	-40	85	°C	-

Note 1: All the above voltages are on the basis of “VSS = 0V” .

Note 2: When this module is used beyond the above absolute maximum ratings, permanent breakage of the module may occur. Also, for normal operations, it is desirable to use this module under the conditions according to Section 3. “Electrical Characteristics” . If this module is used beyond these conditions, malfunctioning of the module can occur and the reliability of the module may deteriorate.

8 . ELECTRICAL CHARACTERISTICS

Characteristics	Symbol	Condition	Min	TYP	Max	Unit
Supply Voltage for Logic	VDD		3.0	3.3	5.0	V
Supply Voltage for I/O Pins	VDDIO		1.6	3.0	3.3	V
Supply Voltage for Display (Generated by Internal DC/DC)	VCC		-	14.0	0	V
High Level Input	VIH	IOU _T = 100μA, 3.3MHz	0.8×V _D D	-	VDDIO	V
Low Level Input	VIL	IOU _T = 100μA, 3.3MHz	0	-	0.2×V _D DIO	V
High Level Output	VOH	IOU _T = 100μA, 3.3MHz	0.9×V _D D	-	VDDIO	V
Low Level Output	VOL	IOU _T = 100μA, 3.3MHz	0	-	0.1×V _D DIO	V

General Electrical Specification

Characteristics	Symbol	Condition	Min	TYP	Max	Unit
Supply Voltage for Logic	VDD		3.0	3.3	5.0	V
Supply Voltage for I/O Pins	VDDIO		1.6	2.8	3.5	V
Driver Supply Voltage	VCC	Note 3	-	14.0	-	V
Operating Current for VDD	IDD	Note 4		0.2	0.6	mA
		Note 5	-	0.2	0.6	mA
Operating Current for VCC	ICC	Note 4	-	8	11	mA
		Note 5	-	13.5	18	mA
Sleep Mode Current for VDD	IDD, SLEEP		-	1	2	μA
Sleep Mode Current for VCC	ICC, SLEEP		-	<2	2	μA

Note 3: Brightness (Lbr) and Driver Supply Voltage (VCC) are subject to the change of the panel characteristics and the customer's request.

Note 4: VDD = 2.8V, VCC = 14V, Software Initial Setting follow Chapter 9.1 “Software Initial Setting”, 50% Display Area Turn on.

Note 5: VDD = 2.8V, VCC = 14V, Software Initial Setting follow Chapter 9.1 “Software Initial Setting”, 100% Display Area Turn on.

9 . Optical Characteristics

Characteristics	Symbol	Condition	Min.	Typ.	Max.	Unit
Brightness(White)	Lbr	With Polarizer (Note 3)	80	100	-	cd/m ^m
C.I.E. (White)	(x)	With Polarizer	0.26	0.30	0.34	cd/m ^m
	(y)		0.30	0.33	0.36	
C.I.E. (Red)	(x)	With Polarizer	0.57	0.61	0.65	
	(y)		0.30	0.34	0.38	
C.I.E. (Green)	(x)	With Polarizer	0.26	0.30	0.34	
	(y)		0.58	0.62	0.66	
C.I.E. (Blue)	(x)	With Polarizer	0.10	0.14	0.18	
	(y)		0.14	0.18	0.22	
Dark Room Contrast	CR		-	>1000:1	-	
Viewing Angle			>160	-	-	degree

* Optical measurement taken at VDD = 2.8V, VCC = 14V, and software configuration follows Sec. 9.1 “Software Initial Setting”.

10. COMMAND TABLE

Fundamental Commands												
D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description	Default
0 0 0	15 A[6:0] B[6:0]	0 * *	0 A ₆ B ₆	0 A ₅ B ₅	1 A ₄ B ₄	0 A ₃ B ₃	1 A ₂ B ₂	0 A ₁ B ₁	1 A ₀ B ₀	Set Column Address	Setup Column start and end address A[6:0] start address from 00d-95d B[6:0] end address from 00d-95d	00d (00h) 95d (5Fh)
0 0 0	75 A[5:0] B[5:0]	0 * *	1 * *	1 A ₅ B ₅	1 A ₄ B ₄	0 A ₃ B ₃	1 A ₂ B ₂	0 A ₁ B ₁	1 A ₀ B ₀	Set Row Address	Setup Row start and end address A[5:0] start address from 00d-63d B[5:0] end address from 00d-63d	00d (00h) 63d (3Fh)
0 0	81 A[7:0]	1 A ₇	0 A ₆	0 A ₅	0 A ₄	0 A ₃	0 A ₂	0 A ₁	1 A ₀	Set Contrast for Color "A"	Set contrast for all color "A" segment (Pins:SA0 – SA95) A[7:0] valid range: 00d to 255d	128d (80h)
0 0	82 A[7:0]	1 A ₇	0 A ₆	0 A ₅	0 A ₄	0 A ₃	0 A ₂	1 A ₁	0 A ₀	Set Contrast for Color "B"	Set contrast for all color "B" segment (Pins:SB0 – SB95). A[7:0] valid range: 00d to 255d	128d (80h)
0 0	83 A[7:0]	1 A ₇	0 A ₆	0 A ₅	0 A ₄	0 A ₃	0 A ₂	1 A ₁	1 A ₀	Set Contrast for Color "C"	Set contrast for all color "C" segment (Pins:SC0 – SC95). A[7:0] valid range: 00d to 255d	128d (80h)
0 0	87 A[3:0]	1 0	0 0	0 0	0 0	0 A ₃	1 A ₂	1 A ₁	1 A ₀	Master Current Control	Set master current attenuation factor A[3:0] from 00d to 15d corresponding to 1/16, 2/16... to 16/16 attenuation.	15d (0Fh)

Fundamental Commands																				
D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description	Default								
0	8A	1	0	0	0	1	0	1	0	Set Second Pre-charge Speed for Color “A”, “B” and “C”	A[7:0]: Set Second Pre-charge Speed Ranges: 0000000b to 1111111b, a higher value of A[7:0] gives a higher Second Pre-charge speed.	A[7:0] of 81h								
0	A[7:0]	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		A[7:0] of 82h									
0	8B	1	0	0	0	1	0	1	1		Note (1) The default values of A[7:0] in 8Ah, A[7:0] in 8Bh and A[7:0] in 8Ch are equal to the contrast values for color A, B and C(refer to commands: 81h, 82h, 83h) respectively. (2) All six bytes (8Ah A[7:0], 8Bh A[7:0] and 8Ch A[7:0]) must be inputted together. For example: the original value is like that <table><tr><td colspan="2">Original value</td></tr><tr><td>8Ah A[7:0]:</td><td>80h</td></tr><tr><td>8Bh A[7:0]:</td><td>80h</td></tr><tr><td>8Ch A[7:0]:</td><td>80h</td></tr></table> If it is wanted to change the value of 8Bh A[7:0] to 75h, then all the following 6 bytes must be inputted: 8Ah,80h, 8Bh,75h, 8Ch,80h.	Original value		8Ah A[7:0]:	80h	8Bh A[7:0]:	80h	8Ch A[7:0]:	80h	A[7:0] of 83h
Original value																				
8Ah A[7:0]:	80h																			
8Bh A[7:0]:	80h																			
8Ch A[7:0]:	80h																			
0	A[7:0]	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀											
0	8C	1	0	0	0	1	1	0	0											
0	A[7:0]	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀											
0	A0	1	0	1	0	0	0	0	0	Remap & Color Depth setting	Set driver remap and color depth A[0]=0, Horizontal address increment A[0]=1, Vertical address increment	A[0]=0								
0	A[7:0]	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		A[1]=0, RAM Column 0 to 95 maps to Pin Seg (SA,SB,SC) 0 to 95 A[1]=1, RAM Column 0 to 95 maps to Pin Seg (SA,SB,SC) 95 to 0	A[1]=0								
											A[2]=0, normal order SA,SB,SC (e.g. RGB) A[2]=1, reverse order SC,SB,SA (e.g. BGR)	A[2]=0								
											A[3]=0, Disable left-right swapping on COM A[3]=1, Set left-right swapping on COM	A[3]=0								
											A[4]=0, Scan from COM 0 to COM [N –1] A[4]=1, Scan from COM [N-1] to COM0. Where N is the multiplex ratio.	A[4]=0								
											A[5]=0, Disable COM Split Odd Even (RESET) A[5]=1, Enable COM Split Odd Even	A[5]=0								
											A[7:6] = 00; 256 color format A[7:6] = 01; 65k color format A[7:6] = 10; 65k color format 2 If 9 / 18 bit mode is selected, color depth will be fixed to 65k regardless of the setting.	A[7:6]=01								
0	A1	1	0	1	0	0	0	0	1	Set Display Start Line	Set display start line register by Row A[5:0]: from 00d to 63d	00d (00h)								
0	A[5:0]	0	0	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀											
0	A2	1	0	1	0	0	0	1	0	Set Display Offset	Set vertical offset by Com A[5:0]: from 00d to 63d	00d (00h)								
0	A[5:0]	0	0	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀											

Fundamental Commands

D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description	Default
0 0 0 0	A4 / A5 / A6 / A7 /	1 	0 	1 	0 	0 	1 	X ₁ 	X ₀ 	Set Display Mode	A4h=Normal Display A5h=Entire Display On, all pixels turn on at GS63 A6h=Entire Display Off, all pixels turn off A7h=Inverse Display	A4h
0 0	A8 A[5:0]	1 0	0 0	1 A ₅	0 A ₄	1 A ₃	0 A ₂	0 A ₁	0 A ₀	Set Multiplex Ratio	Set MUX ratio to N+1 Mux N = A[5:0] from 15d to 63d A[5:0] from 00d to 14d are invalid entry	63d (3Fh)
0 0 0 0 0 0	AB A[7:0] B[7:0] C[7:0] D[7:0] E[4:0]	1 A ₇ B ₇ C ₇ D ₇ 0	0 A ₆ B ₆ C ₆ D ₆ 0	1 A ₅ B ₅ C ₅ D ₅ 0	0 A ₄ B ₄ C ₄ D ₄ E ₄	1 A ₃ B ₃ C ₃ D ₃ E ₃	0 A ₂ B ₂ C ₂ D ₂ E ₂	1 A ₁ B ₁ C ₁ D ₁ E ₁	1 A ₀ B ₀ C ₀ D ₀ E ₀	Dim Mode Setting	Configure dim mode setting A[7:0] = Reserved. (Set as 00h) B[7:0] = Contrast setting for Color A, valid range 0 to 255d. C[7:0] = Contrast setting for Color B, valid range 0 to 255d. D[7:0] = Contrast setting for Color C, valid range 0 to 255d. E[4:0] = Precharge voltage setting, valid range 0 to 31d.	\
0 0	AD A[0]	1 1	0 0	1 0	0 0	1 1	1 1	0 1	1 A ₀	Set Master Configuration	A[0]=0b, Select external V _{CC} supply A[0]=1b, Reserved (RESET) Note (1) Bit A[0] must be set to 0b after RESET. (2) The setting will be activated after issuing Set Display ON command (AFh)	A[0] = 1
0	AC AE AF	1	0	1	0	1	1	A ₁	A ₀	Set Display On/Off	ACH = Display ON in dim mode AEh = Display off (sleep mode) AFh = Display on in normal mode	AEh
0 0	B0 A[7:0]	1 A ₇	0 A ₆	1 A ₅	1 A ₄	0 A ₃	0 A ₂	0 A ₁	0 A ₀	Power Save Mode	A[7:0]=1Ah, Enable Power save mode (RESET) A[7:0]=0Bh, Disable Power save mode	1Ah
0 0	B1 A[7:0]	1 A ₇	0 A ₆	1 A ₅	1 A ₄	0 A ₃	0 A ₂	0 A ₁	1 A ₀	Phase 1 and 2 period adjustment	A[3:0] Phase 1 period in N DCLK. 1~15 DCLK allowed. A[7:4] Phase 2 period in N DCLK. 1~15 DCLK allowed	74h
0 0	B3 A[7:0]	1 A ₇	0 A ₆	1 A ₅	1 A ₄	0 A ₃	0 A ₂	1 A ₁	1 A ₀	Display Clock Divider / Oscillator Frequency	A[3:0]: Define the divide ratio (D) of the display clocks (DCLK): Divide ratio (D) = A[3:0] + 1 (i.e., 1 to 16) A[7:4] Fosc frequency. Frequency increases as setting value increases	D0h

Fundamental Commands																														
D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description	Default																		
0	B8	1	0	1	1	1	0	0	0	Set Gray Scale Table	These 32 parameters define pulse widths of GS1 to GS63 in terms of DCLK A[6:0]: Pulse width for GS1, RESET=01d B[6:0]: Pulse width for GS3, RESET=05d C[6:0]: Pulse width for GS5, RESET=09d ... AE[6:0]: Pulse width for GS61, RESET=121d AF[6:0]: Pulse width for GS63, RESET=125d Note: (1) GS0 has no pre-charge and current drive stages. (2) GS2, GS4...GS62 are derived by Pn = (Pn-1+Pn+1)/2 (3) Pn will be truncated to integer if it is with decimal point. (4) Pn+1 should always be set to larger than Pn-1 (5) Max pulse width is 125	\																		
0	A[6:0]	*	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																					
0	B[6:0]	*	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																					
0	C[6:0]	*	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀																					
0	...	...	...	...	...	...	...	...	...																					
0	AE[6:0]	*	AE ₆	AE ₅	AE ₄	AE ₃	AE ₂	AE ₁	AE ₀																					
0	AF[6:0]	*	AF ₆	AF ₅	AF ₄	AF ₃	AF ₂	AF ₁	AF ₀																					
0	B9	1	0	1	1	1	0	0	1	Enable Linear Gray Scale Table	Reset built in gray scale table (Linear) Pulse width for GS1 = 1d; Pulse width for GS2 = 3d; Pulse width for GS3 = 5d; ... Pulse width for GS61 = 121d; Pulse width for GS62 = 123d; Pulse width for GS63 = 125d.	\																		
0	BB	1	0	1	1	1	0	1	1	Set Pre-charge level	Set pre-charge voltage level. All three color share the same pre-charge voltage. <table><tr><th>A[5:1]</th><th>Hex code</th><th>pre-charge voltage</th></tr><tr><td>00000</td><td>00h</td><td>0.10 x V_{CC}</td></tr><tr><td>:</td><td>:</td><td>:</td></tr><tr><td>11111</td><td>3Eh</td><td>0.50 x V_{CC}</td></tr></table> Refer to Figure 31 for the details setting of A[5:1].	A[5:1]	Hex code	pre-charge voltage	00000	00h	0.10 x V _{CC}	:	:	:	11111	3Eh	0.50 x V _{CC}	3Eh						
A[5:1]	Hex code	pre-charge voltage																												
00000	00h	0.10 x V _{CC}																												
:	:	:																												
11111	3Eh	0.50 x V _{CC}																												
0	A[5:0]	0	0	A ₅	A ₄	A ₃	A ₂	A ₁	0																					
0	BC-BD	1	0	1	1	1	1	0	X ₀	NOP	Command for No operation	\																		
0	BE	1	0	1	1	1	1	1	0	Set V _{COMH}	Set COM deselect voltage level (V _{COMH}) <table><tr><th>A[5:1]</th><th>Hex code</th><th>V_{COMH}</th></tr><tr><td>00000</td><td>00h</td><td>0.44 x V_{CC}</td></tr><tr><td>01000</td><td>10h</td><td>0.52 x V_{CC}</td></tr><tr><td>10000</td><td>20h</td><td>0.61 x V_{CC}</td></tr><tr><td>11000</td><td>30h</td><td>0.71 x V_{CC}</td></tr><tr><td>11111</td><td>3Eh</td><td>0.83 x V_{CC}</td></tr></table>	A[5:1]	Hex code	V _{COMH}	00000	00h	0.44 x V _{CC}	01000	10h	0.52 x V _{CC}	10000	20h	0.61 x V _{CC}	11000	30h	0.71 x V _{CC}	11111	3Eh	0.83 x V _{CC}	3Eh
A[5:1]	Hex code	V _{COMH}																												
00000	00h	0.44 x V _{CC}																												
01000	10h	0.52 x V _{CC}																												
10000	20h	0.61 x V _{CC}																												
11000	30h	0.71 x V _{CC}																												
11111	3Eh	0.83 x V _{CC}																												
0	A[5:1]	0	0	A ₅	A ₄	A ₃	A ₂	A ₁	0																					
0	E3	1	1	1	0	0	0	1	1	NOP	Command for No operation	\																		
0	FD	1	1	1	1	1	1	0	1	Set Command Lock	A[2]: MCU protection status A[2] = 0b, Unlock OLED driver IC MCU interface from entering command [reset] A[2] = 1b, Lock OLED driver IC MCU interface from entering command Note (1) The locked OLED driver IC MCU interface prohibits all commands and memory access except the FDh command.	12h																		
0	A[2]	0	0	0	1	0	A ₂	1	0																					

Graphic Acceleration Commands

D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	21	0	0	1	0	0	0	0	1	Draw Line	A[6:0]: Column Address of Start
0	A[6:0]	*	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		B[5:0]: Row Address of Start
0	B[5:0]	*	*	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		C[6:0]: Column Address of End
0	C[6:0]	*	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		D[5:0]: Row Address of End
0	D[5:0]	*	*	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		E[5:1]: Color C of the line
0	E[5:1]	*	*	E ₅	E ₄	E ₃	E ₂	E ₁	*		F[5:0]: Color B of the line
0	F[5:0]	*	*	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀		G[5:1]: Color A of the line
0	G[5:1]	*	*	G ₅	G ₄	G ₃	G ₂	G ₁	*		
0	22	0	0	1	0	0	0	1	0	Drawing Rectangle	A[6:0]: Column Address of Start
0	A[6:0]	*	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		B[5:0]: Row Address of Start
0	B[5:0]	*	*	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		C[6:0]: Column Address of End
0	C[6:0]	*	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		D[5:0]: Row Address of End
0	D[5:0]	*	*	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		E[5:1]: Color C of the line
0	E[5:1]	*	*	E ₅	E ₄	E ₃	E ₂	E ₁	*		F[5:0]: Color B of the line
0	F[5:0]	*	*	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀		G[5:1]: Color A of the line
0	G[5:1]	*	*	G ₅	G ₄	G ₃	G ₂	G ₁	*		H[5:1]: Color C of the fill area
0	H[5:1]	*	*	H ₅	H ₄	H ₃	H ₂	H ₁	*		I[5:0]: Color B of the fill area
0	I[5:0]	*	*	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀		J[5:1]: Color A of the fill area
0	J[5:1]	*	*	J ₅	J ₄	J ₃	J ₂	J ₁	*		
0	23	0	0	1	0	0	0	1	1	Copy	A[6:0]: Column Address of Start
0	A[6:0]	*	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		B[5:0]: Row Address of Start
0	B[5:0]	*	*	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		C[6:0]: Column Address of End
0	C[6:0]	*	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		D[5:0]: Row Address of End
0	D[5:0]	*	*	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		E[6:0]: Column Address of New Start
0	E[6:0]	*	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀		F[5:0]: Row Address of New Start
0	F[5:0]	*	*	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀		
0	24	0	0	1	0	0	1	0	0	Dim Window	A[6:0]: Column Address of Start
0	A[6:0]	*	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		B[5:0]: Row Address of Start
0	B[5:0]	*	*	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		C[6:0]: Column Address of End
0	C[6:0]	*	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		D[5:0]: Row Address of End
0	D[5:0]	*	*	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		The effect of dim window: GS15~GS0 no change GS19~GS16 become GS4 GS23~GS20 become GS5 ... GS63~GS60 become GS15
0	25	0	0	1	0	0	1	0	1	Clear Window	A[6:0]: Column Address of Start
0	A[6:0]	*	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		B[5:0]: Row Address of Start
0	B[5:0]	*	*	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		C[6:0]: Column Address of End
0	C[6:0]	*	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		D[5:0]: Row Address of End
0	D[5:0]	*	*	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		
0	26	0	0	1	0	0	1	1	0	Fill Enable / Disable	A0 0 : Disable Fill for Draw Rectangle Command (RESET) 1 : Enable Fill for Draw Rectangle Command
0	A[4:0]	*	*	*	A ₄	0	0	0	A ₀		A[3:1] 000: Reserved values A4 0 : Disable reverse copy (RESET) 1 : Enable reverse during copy command.

Graphic Acceleration Commands											
D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	27	0	0	1	0	0	1	1	1	Continuous Horizontal & Vertical Scrolling Setup	A[6:0]: Set number of column as horizontal scroll offset Range: 0d-95d (no horizontal scroll if equals to 0)
0	A[6:0]	*	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		B[5:0]: Define start row address
0	B[5:0]	*	*	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		C[6:0]: Set number of rows to be horizontal scrolled B[5:0]+C[6:0] <=64
0	C[6:0]	*	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		D[5:0]: Set number of row as vertical scroll offset Range: 0d-63d (no vertical scroll if equals to 0)
0	D[5:0]	*	*	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		E[1:0]: Set time interval between each scroll step 00b 6 frames 01b 10 frames 10b 100 frames 11b 200 frames
0	E[1:0]	*	*	*	*	*	*	E ₁	E ₀		Note: (1) Vertical scroll is run with 64MUX setting only (2) The parameters should not be changed after scrolling is activated
0	2E	0	0	1	0	1	1	1	0	Deactivate scrolling	This command deactivates the scrolling action. Note (1) After sending 2Eh command to deactivate the scrolling action, the ram data needs to be rewritten.
0	2F	0	0	1	0	1	1	1	1	Activate scrolling	This command activates the scrolling function according to the setting done by Continuous Horizontal & Vertical Scrolling Setup command 27h.

For detailed instruction information, see SSD1331 datasheet .

11 . MPU Interface

Table 21 - Serial Interface Timing Characteristics

($V_{DD} - V_{SS} = 2.4V$ to $3.5V$, $V_{DDIO} = 2.4V$ to V_{DD} , $T_A = -40$ to $85^{\circ}C$)

Symbol	Parameter	Min	Typ	Max	Unit
t_{cycle}	Clock Cycle Time	150	-	-	ns
t_{AS}	Address Setup Time	40	-	-	ns
t_{AH}	Address Hold Time	40	-	-	ns
t_{CSS}	Chip Select Setup Time	75	-	-	ns
t_{CSH}	Chip Select Hold Time	60	-	-	ns
t_{DSW}	Write Data Setup Time	40	-	-	ns
t_{DHW}	Write Data Hold Time	40	-	-	ns
t_{CLKL}	Clock Low Time	75	-	-	ns
t_{CLKH}	Clock High Time	75	-	-	ns
t_R	Rise Time	-	-	15	ns
t_F	Fall Time	-	-	15	ns

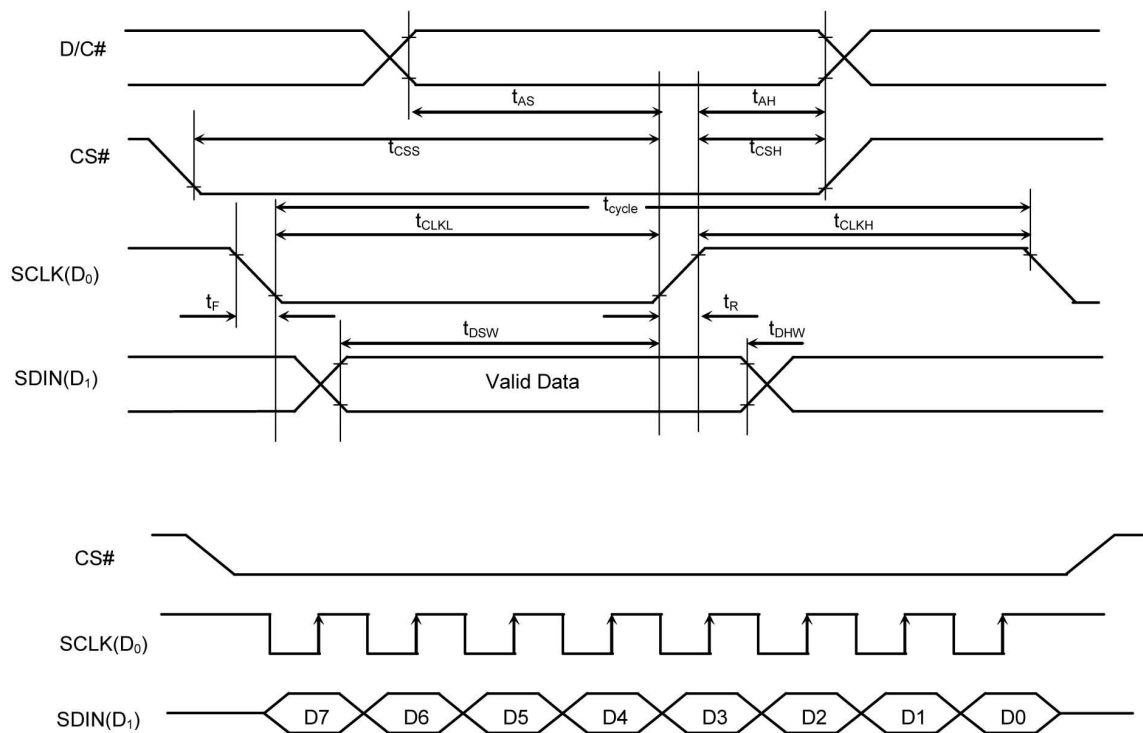


Figure 40 - Serial interface characteristics

12. Graphic Display Data RAM (GDDRAM)

GDDRAM structure

The GDDRAM is a bit mapped static RAM holding the pattern to be displayed. The RAM size is $96 \times 64 \times 16$ bits.

For mechanical flexibility, re-mapping on both Segment and Common outputs can be selected by software. For vertical scrolling of the display, an internal register storing display start line can be set to control the portion of the RAM data to be mapped to the display.

Each pixel has 16-bit data. Three sub-pixels for color A, B and C have 6 bits, 5 bits and 6 bits respectively. The arrangement of data pixel in graphic display data RAM is shown below.

Figure 10 - 65k Color Depth Graphic Display Data RAM Structure

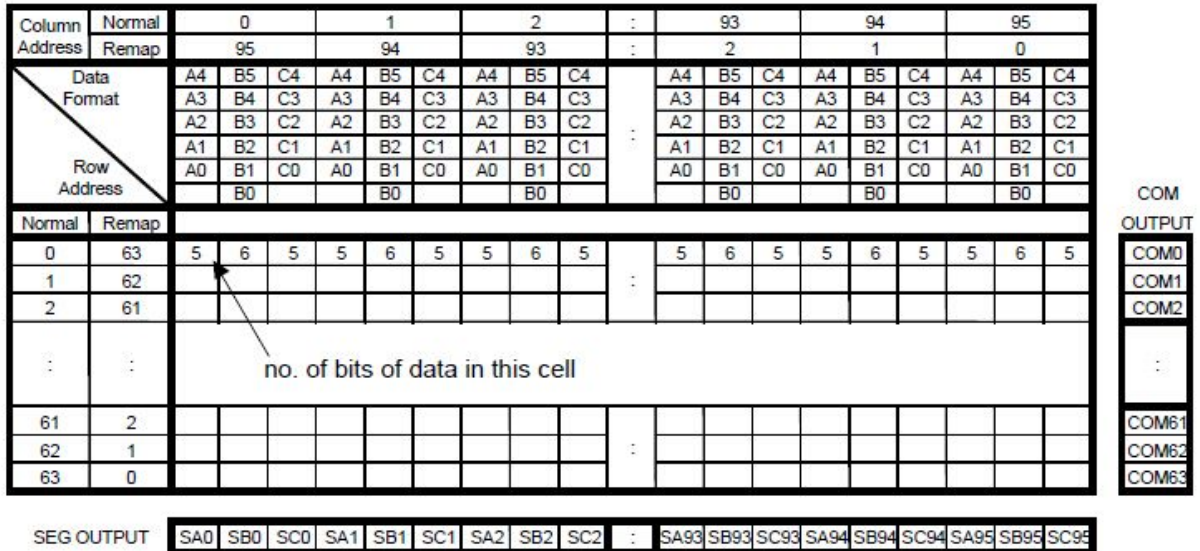


Table 9 - Data bus usage under different bus width and color depth mode

Bus width	Color Depth	Input order	Data bus															
			D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
8 bits	256		X	X	X	X	X	X	X	X	C ₄	C ₃	C ₂	B ₅	B ₄	B ₃	A ₄	A ₃
8 bits	65k format 1	1st	X	X	X	X	X	X	X	X	C ₄	C ₃	C ₂	C ₁	C ₀	B ₅	B ₄	B ₃
		2nd	X	X	X	X	X	X	X	X	B ₂	B ₁	B ₀	A ₄	A ₃	A ₂	A ₁	A ₀
8 bits	65k format 2	1st	X	X	X	X	X	X	X	X	X	X	C ₄	C ₃	C ₂	C ₁	C ₀	X
		2nd	X	X	X	X	X	X	X	X	X	X	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀
		3rd	X	X	X	X	X	X	X	X	X	X	A ₄	A ₃	A ₂	A ₁	A ₀	X
16 bits	65k		C ₄	C ₃	C ₂	C ₁	C ₀	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀	A ₄	A ₃	A ₂	A ₁	A ₀
9 bits	65k	1st	X	X	X	X	X	X	X	C ₄	C ₃	C ₂	C ₁	C ₀	X	B ₅	B ₄	B ₃
		2nd	X	X	X	X	X	X	X	B ₂	B ₁	B ₀	A ₄	A ₃	A ₂	A ₁	A ₀	X

13. Gray Scale Decoder

The gray scale effect is generated by controlling the pulse width of segment drivers in current drive phase. The gray scale table stores the corresponding pulse widths of the 63 gray scale levels (GS0~GS63). The wider the pulse width, the brighter the pixel will be. A single gray scale table supports all the three colors A, B and C. The pulse widths can be set by software commands.

As shown in Figure 12, color B sub-pixel RAM data has 6 bits, represent the 64 gray scale levels from GS0 to GS63. color A and color C sub-pixel RAM data has only 5 bits, represent 32 gray scale levels from GS0, GS2, ..., GS62.

Figure 12 - Relation between GDRAM content and gray scale table entry for three colors in 65K color mode

Color A, C RAM data (5 bits)	Color B RAM data (6 bits)	Gray Scale	Default pulse width of GS[0:63] in terms of DCLK
00000	000000	GS0	0
-	000001	GS1	1
00001	000010	GS2	3
-	000011	GS3	5
00010	000100	GS4	7
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮
11110	111100	GS60	119
-	111101	GS61	121
11111	111110	GS62	123
-	111111	GS63	125

The duration of different GS are programmable.

14 . DESIGN AND HANDING PRECAUTION

14.1 The LCD panel is made by glass. Any mechanical shock (eg. Dropping form high place) will damage the LCD module. Do not add excessive force on the surface of the display, which may cause the Display color change abnormally.

14.2 The polarizer on the LCD is easily get scratched. If possible, do not remove the LCD protective film until the last step of installation.

14.3 Never attempt to disassemble or rework the LCD module.

14.4 Only Clean the LCD with Isopropyl Alcohol or Ethyl Alcohol. Other solvents (eg. water) may damage the LCD.

14.5 When mounting the LCD module, make sure that it is free form twisting, warping and distortion.

14.6 Ensure to provide enough space(with cushion) between case and LCD panel to prevent external force adding on it, or it may cause damage to the LCD or degrade the display result

14.7 Only hold the LCD module by its side. Never hold LCD module by add force on the heat seal or TAB.

14.8 Never add force to component of the LCD module. It may cause invisible damage or degrade of the reliability.

14.9 LCD module could be easily damaged by static electricity. Be careful to maintain an optimum anti-static work environment to protect the LCD module.

14.10 When peeling of the protective film form LCD, static charge may cause abnormal display pattern. It is normal and will resume to normal in a short while.

14.11 Take care and prevent get hurt by the LCD panel edge.

- 14.12 Never operate the LCD module exceed the absolute maximum ratings.
- 14.13 Keep the signal line as short as possible to prevent noisy signal applying to LCD module.
- 14.14 Never apply signal to the LCD module without power supply.
- 14.15 IC chip (eg. TAB or COG) is sensitive to the light. Strong lighting environment could possibly cause malfunction. Light sealing structure casing is recommend.
- 14.16 LCD module reliability may be reduced by temperature shock.
- 14.17 When storing the LCD module, avoid exposure to the direct sunlight, high humidity, high temperature or low temperature. They may damage or degrade the LCD module